Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_random_replay.
- Best TSPLIB holdout gap: tsplib_random_replay.
- Best synthetic holdout gap: tsplib_no_replay.

Run Metadata

- run_name: run_20260513_143211_l
- started_at_local: 2026-05-13 14:32:11
- finished_at_local: 2026-05-13 14:42:57
- duration_hhmm: 00:11
- duration_seconds: 646.273
- seed_offset: 11000
- replicate_label: l
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.16877	0.0	0.107399	0.0	0.76	0.0
tsplib_random_replay	random	score_only	False	0.097675	0.0	0.062157	0.747293	0.76	0.024494
tsplib_failure_replay	failure	score_only	False	0.111105	0.0	0.070703	0.788828	0.76	0.16106
tsplib_failure_replay_compression	failure	novelty_gate	True	0.176277	0.010343	0.115937	0.0	0.76	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.16877, synthetic 0.0, combined 0.107399.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.16877 across 7 instances; family means: ch=0.186649, kroD=0.149291, pcb=0.235141, pr=0.246091, rd=0.122756, st=0.054815.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3445, "family": "a", "name": "a280", "optimality_gap": 0.335789}, {"best_known_cost": 14379, "cost": 18952, "family": "lin", "name": "lin105", "optimality_gap": 0.318033}, {"best_known_cost": 629, "cost": 738, "family": "eil", "name": "eil101", "optimality_gap": 0.173291}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.097675, synthetic 0.0, combined 0.062157.
- Accepted-epoch count 3, mean accepted code novelty 0.747293, and final complexity 0.76.
- Adaptation efficiency 0.024494 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.097675 across 7 instances; family means: ch=0.183951, kroD=0.052221, pcb=0.162275, pr=0.036113, rd=0.032617, st=0.032593.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3428, "family": "a", "name": "a280", "optimality_gap": 0.329197}, {"best_known_cost": 629, "cost": 765, "family": "eil", "name": "eil101", "optimality_gap": 0.216216}, {"best_known_cost": 14379, "cost": 17241, "family": "lin", "name": "lin105", "optimality_gap": 0.19904}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.111105, synthetic 0.0, combined 0.070703.
- Accepted-epoch count 2, mean accepted code novelty 0.788828, and final complexity 0.76.
- Adaptation efficiency 0.16106 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.111105 across 7 instances; family means: ch=0.098865, kroD=0.070583, pcb=0.249439, pr=0.112871, rd=0.071555, st=0.075556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3467, "family": "a", "name": "a280", "optimality_gap": 0.34432}, {"best_known_cost": 7542, "cost": 9813, "family": "berlin", "name": "berlin52", "optimality_gap": 0.301114}, {"best_known_cost": 629, "cost": 753, "family": "eil", "name": "eil101", "optimality_gap": 0.197138}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.176277, synthetic 0.010343, combined 0.115937.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.176277 across 7 instances; family means: ch=0.168312, kroD=0.250305, pcb=0.216846, pr=0.240627, rd=0.164349, st=0.025185.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19083, "family": "lin", "name": "lin105", "optimality_gap": 0.327144}, {"best_known_cost": 2579, "cost": 3332, "family": "a", "name": "a280", "optimality_gap": 0.291974}, {"best_known_cost": 629, "cost": 685, "family": "eil", "name": "eil101", "optimality_gap": 0.08903}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Results

Condition	Replay Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (mean)	Notes on Transfer & Novelty
tsplib_no_replay	None	0.16877	0.0	0.107399	0.0	Baseline with no replay; moderate transfer gap; no novelty.
tsplib_random_replay	Random	**0.097675**	0.0	**0.062157**	0.747293	Best transfer performance on TSPLIB and best holdout; higher code novelty but no evidence that novelty correlates with better algorithmic behavior.
tsplib_failure_replay	Failure	0.111105	0.0	0.070703	0.788828	Slightly worse transfer than random replay, despite highest adaptation efficiency and code novelty.
tsplib_failure_replay_compression	Failure + Compression	0.176277	0.010343	0.115937	0.0	Worst transfer performance; compression pressure present; no code novelty observed.

Key Interpretations

- **Transfer Metrics (Primary Evidence):**

The **random replay condition** demonstrates the best transfer performance, reflected in the lowest TSPLIB optimality gap (0.097675) and the lowest mean transfer gap (0.062157). The **no replay condition** performs worse in transfer (TSPLIB gap 0.16877; transfer gap 0.107399), while both failure replay conditions perform intermediately to poorly.

- **Synthetic Holdout Gap:**

Only the compression-aware failure replay condition shows a notable synthetic holdout gap (0.0103), indicating slightly worse generalization there. Other replay modes achieve zero synthetic holdout gap, indicating good synthetic domain transfer.

- **Code Novelty vs Transfer:**

The random replay and failure replay modes show high mean code novelty (~0.75-0.79), while no replay and compression-aware failure replay show zero. However, better transfer (random replay) is associated with **higher code novelty** than no replay, which contradicts a simplistic assumption that novelty should be low to improve transfer. The compression-aware failure replay lacks code novelty but performs worst on transfer, suggesting code novelty alone does not drive generalization.

- **Replay Mode Implications:**
- **No Replay:** Baseline with moderate transfer and no code novelty.
- **Random Replay:** Best transfer with higher code novelty; implies replay diversity boosts transfer but lowers code novelty.
- **Failure Replay:** Moderate transfer, highest adaptation efficiency, high code novelty but no clear transfer advantage over random replay.
- **Failure Replay + Compression:** Compression pressure seems to impair transfer and induce some synthetic gap; no code novelty present.

- **Complexity and Behavior Profiles:**

All conditions have identical final complexity (0.76), so complexity is not a factor in transfer differences. Behavior profile shifts only reported in compression-aware failure replay (clustered_constructor), but without corresponding transfer advantage.

Conclusion

- The **random replay condition yields the best transfer to held-out TSPLIB and synthetic instances**, supported by the lowest mean optimality gaps, despite increased code novelty compared to no replay.
- Novelty metrics should be interpreted cautiously; higher code novelty in random replay does not imply lack of improvement — rather, transfer improves while novelty increases, showing novelty and transfer are not inversely related here.
- Failure replay conditions, while featuring strong adaptation efficiency and code novelty, do not outperform random replay in transfer metrics.
- Compression-aware failure replay degrades transfer and generalization despite replay use, with no code novelty, emphasizing that compression pressure can harm performance.
- Overall, **random replay is the recommended replay strategy for optimal transfer, with no evidence that code novelty loss is necessary to gain this benefit.**